A non-transitive 6-edge-coloured tournament on 9 vertices with a shorter longest colour-avoiding path than any transitive one: an answer to Problem 5.1 of Fox, Sudakov and Wigderson

github.com/graph-theory-AI

Machine-generated note (see provenance box)

9 September 2026

Provenance

The mathematics in this note was produced without human intervention, in three stages. (1) The construction and proof were found by the language model `gpt-5.6-sol` in a single autonomous pass on 2026-08-31, as part of a large sweep over open problems catalogued from arXiv (catalog id `2512.10438__00`; original writeup in `attacks/2512.10438__00/output.md`). (2) An independent adversarial referee agent (`claude-fable-5`) re-derived every step, compared the problem statement and definitions verbatim with the source paper, re-encoded the tournament and brute-forced all its directed paths, and verified the transitive benchmark $f_{6,5}(9) = 8$ exhaustively over all 6^8 colourings of the consecutive edges; its verdict was CONFIRMED. Its report is reproduced verbatim in Appendix B. (3) On 2026-09-09 the writeup was rewritten into the present note by `claude-fable-5.1`: the text was reorganised with full proofs in Appendix A, the colouring was tabulated edge by edge (the writeup left the "remaining" edges to colour 1 implicitly), and two clarifications taken from the referee's report were added and are marked as such (the count of 21 eight-vertex paths, and why the unspecified edge colours are irrelevant). No mathematical content was changed. **No human mathematician has checked this argument.** We circulate it to obtain exactly that: is the proof correct, and is the result new?

Abstract

Fox, Sudakov and Wigderson study, for a q-edge-coloured tournament on N vertices, the longest directed path whose edges miss at least one colour, and denote by $f_{q,q-1}(N)$ the minimum length (in vertices) of such a path over all q-edge-colourings of the transitive N-vertex tournament. Their Problem 5.1 asks whether, for some $q \geq 4$ and N, a *non-transitive* q-edge-coloured tournament on N vertices can have a strictly shorter longest colour-avoiding path than $f_{q,q-1}(N)$. We answer this affirmatively with $(q, N) = (6, 9)$: we show $f_{6,5}(9) = 8$ and exhibit a 6-edge-coloured tournament on 9 vertices containing one directed triangle whose longest colour-avoiding path has 7 vertices.

1 Introduction

Throughout, following [1], the *length* of a directed path is the number of vertices it comprises. Let T be a tournament and χ a colouring of its edges with q colours (not necessarily all used). A directed path is *colour-avoiding* if its edges use at most $q-1$ of the q colours, i.e. miss at least one colour. Write $p(\chi)$ for the maximum length of a colour-avoiding directed path. Following [1] let $g_{q,q-1}(N)$ be the minimum of $p(\chi)$ over all q-edge-coloured N-vertex tournaments, and

$$f_{q,q-1}(N) = \min_{\chi} \; p(\chi),$$

the minimum over all q-edge-colourings χ of the *transitive* tournament on N vertices. Trivially $g_{q,q-1}(N) \leq f_{q,q-1}(N)$. For $q = 3$ the inequality can be strict, by a construction of Hamaker and

Stein discussed in [1]; for $q \geq 4$ the authors show (their Proposition 1.5) that the vector-sequence constructions which produce such examples cannot work, and they ask:

Problem 1.1 ([1, Problem 5.1])**.** For some integers $q \geq 4$ and N, does there exist a non-transitive q-edge-coloured N-vertex tournament whose longest colour-avoiding path has strictly fewer than $f_{q,q-1}(N)$ vertices?

The authors write that they do not expect $f_{q,q-1}(N) = g_{q,q-1}(N)$ in general, so an affirmative answer is the anticipated direction. We give one.

Theorem 1.2 (Main theorem)**.** $f_{6,5}(9) = 8$*, and there is a* 6*-edge-coloured non-transitive tournament* T *on* 9 *vertices, described explicitly in Section 3, whose longest colour-avoiding directed path has exactly* 7 *vertices.*

Corollary 1.3. *Problem 1.1 has an affirmative answer, with* $(q, N) = (6, 9)$*: the tournament* T *of Theorem 1.2 satisfies* $p(T) = 7 < 8 = f_{6,5}(9)$*.*

Proof. Immediate from Theorem 1.2, since T is non-transitive (it contains a directed triangle). □

Idea of proof. Theorem 1.2 is the conjunction of Lemma 2.1 ($f_{6,5}(9) = 8$) and Lemma 3.1 ($p(T) = 7$). For the benchmark, the 8-vertex paths of the transitive tournament $v_1 \prec \cdots \prec v_9$ are the nine vertex-deleted increasing sequences; if none of them were colour-avoiding, a counting argument on the word of colours of the eight consecutive edges $v_i v_{i+1}$ would force four colours to occur exactly once, at pairwise non-consecutive interior positions, which is impossible in six positions. For the construction, T consists of a transitive triple L, a directed triangle C and a transitive triple R with all edges pointing $L \to C \to R$ and $L \to R$; every 8-vertex path visits the blocks in this order, and the colouring is arranged so that each of the 21 such paths uses all six colours, while the 7-vertex path $\ell_1 \ell_2 \ell_3 x_0 x_1 x_2 r_1$ avoids colour 4.

2 The transitive benchmark

Lemma 2.1. $f_{6,5}(9) = 8$.

Idea of proof. Lower bound. Let $v_1, \ldots, v_9$ be the transitive order and $a_i = \chi(v_i v_{i+1})$. If no 8-vertex path is colour-avoiding, then $a_1 \cdots a_7$ and $a_2 \cdots a_8$ each contain all six colours, so at least four colours occur exactly once in $a_1 \cdots a_8$ (*singleton positions*); positions 1 and 8 cannot be singleton, and two singleton positions cannot be consecutive (deleting the vertex between them removes both colours and adds a single chord, which can restore at most one). Four pairwise non-consecutive positions do not fit in $\{2, \ldots, 7\}$. *Upper bound.* Colour the consecutive edges $1, 2, 3, 4, 5, 6, 1, 2$; the unique 9-vertex path uses all six colours. *[full proof]*

3 The construction

Let the vertex set be $L \cup C \cup R$ with

$$L = \{\ell_1, \ell_2, \ell_3\}, \qquad C = \{x_0, x_1, x_2\}, \qquad R = \{r_1, r_2, r_3\},$$

indices of the x_i being read modulo 3. Orient L transitively as $\ell_1 \to \ell_2 \to \ell_3$, R transitively as $r_1 \to r_2 \to r_3$, C as the directed triangle $x_0 \to x_1 \to x_2 \to x_0$, and every edge between blocks from L to C, from C to R, and from L to R. This is a tournament on 9 vertices, non-transitive because C is a directed triangle.

Put $d_0 = 2$, $d_1 = 5$, $d_2 = 6$ (indices modulo 3) and $D = \{d_0, d_1, d_2\} = \{2, 5, 6\}$. The colouring χ is given in Table 1; in words: $\chi(\ell_1\ell_2) = 1$, $\chi(\ell_2\ell_3) = \chi(\ell_1\ell_3) = 3$, $\chi(r_1r_2) = \chi(r_1r_3) = 4$, $\chi(r_2r_3) = 1$, $\chi(x_ix_{i+1}) = d_i$, $\chi(\ell_3x_i) = d_{i-1}$, $\chi(x_ir_1) = d_i$, $\chi(\ell_2x_i) = 3$, $\chi(x_ir_2) = 4$ for all $i \in \mathbb{Z}/3\mathbb{Z}$, and every remaining edge (the three edges ℓ_1x_i, the three edges x_ir_3, and the nine edges from L to R) has colour 1. All six colours occur.

edges	colour	comment
$\ell_1\ell_2$	1	
$\ell_2\ell_3$, $\ell_1\ell_3$	3	
r_1r_2, r_1r_3	4	
r_2r_3	1	
x_0x_1, x_1x_2, x_2x_0	2, 5, 6	$\chi(x_ix_{i+1}) = d_i$
ℓ_3x_0, ℓ_3x_1, ℓ_3x_2	6, 2, 5	$\chi(\ell_3x_i) = d_{i-1}$
x_0r_1, x_1r_1, x_2r_1	2, 5, 6	$\chi(x_ir_1) = d_i$
ℓ_2x_0, ℓ_2x_1, ℓ_2x_2	3	
x_0r_2, x_1r_2, x_2r_2	4	
ℓ_1x_0, ℓ_1x_1, ℓ_1x_2	1	not used by any path on ≥ 8 vertices
x_0r_3, x_1r_3, x_2r_3	1	not used by any path on ≥ 8 vertices
ℓ_ar_b $(a, b \in \{1, 2, 3\})$	1	not used by any path on ≥ 8 vertices

Table 1: The 6-edge-colouring χ of T; all 36 edges are listed (orientation: ℓ's $\to x$'s $\to r$'s, $\ell_1 \to \ell_2 \to \ell_3$, $r_1 \to r_2 \to r_3$, $x_0 \to x_1 \to x_2 \to x_0$).

Lemma 3.1. *Every directed path of T on 8 or more vertices uses all six colours, and the path $\ell_1\ell_2\ell_3x_0x_1x_2r_1$ has edge colours* $1, 3, 6, 2, 5, 6$. *Hence* $p(T) = 7$.

Idea of proof. Since all inter-block edges point $L \to C \to R$, an 8-vertex path omits exactly one vertex and visits L, C, R in this order, the L- and R-parts in their transitive order and the C-part as a cyclic rotation x_i, x_{i+1}, x_{i+2} or as a single edge x_ix_{i+1}. With boundary vertices ℓ_3 and r_1 present, the entry edge, the triangle edges and the exit edge carry the colours $d_{i-1}, d_i, d_{i+1}, d_{i+2}$ (or d_{i-1}, d_i, d_{i+1}), covering D; a case analysis over the omitted vertex shows that the colours $1, 3, 4$ are always present too, using the chords $\ell_1\ell_3$, r_1r_3 and the alternative boundary edges ℓ_2x_i, x_ir_2 when ℓ_2, r_2, ℓ_3 or r_1 is omitted. A 9-vertex path contains an 8-vertex subpath. *[full proof]*

Proof of Theorem 1.2. Combine Lemma 2.1 and Lemma 3.1. □

4 Remarks

Remark 4.1 (What is and is not answered). Problem 1.1 asks for *some* (q, N), and is resolved by the single instance $(6, 9)$ with a gap of one vertex. The asymptotic question the authors of [1] also care about, whether $g_{q,q-1}(N)$ is asymptotically smaller than $f_{q,q-1}(N)$ for $q \geq 4$ (for $q = 3$ the gap is a power of N), is not addressed.

Remark 4.2 (Connectivity). T is non-transitive but not strongly connected: its strong components are $\ell_1, \ell_2, \ell_3, \{x_0, x_1, x_2\}, r_1, r_2, r_3$ in this order. Problem 1.1 imposes no connectivity requirement.

Remark 4.3 (No conflict with Proposition 1.5 of the source paper). Proposition 1.5 of [1] concerns the vector setting ($(q-1)$-comparable sets versus $(q-1)$-increasing sequences) and rules out vector-based constructions for $q \geq 4$; the implication in [1] from vector constructions to tournament bounds is one-directional, so a direct tournament construction such as T is not constrained by it. (Observation taken from the referee's report, Appendix B.)

Remark 4.4 (The unspecified edges). The edges assigned colour 1 "by default" in Table 1 never lie on a directed path with 8 or more vertices: such a path uses only the edges listed in the case analysis of Lemma 3.1. So the conclusion $p(T) \leq 7$ is independent of how they are coloured; the referee's enumeration of all 21 eight-vertex paths confirms this (Appendix B). The value $p(T) = 7$ was checked by exhaustive search over all directed paths of T with the colouring exactly as tabulated.

Remark 4.5 (Computational verification). The referee (Appendix B) re-encoded T and found: 36 edges, one per pair; a directed triangle; all six colours used; exactly 21 directed paths on 8 vertices, each using all six colours; longest colour-avoiding path of 7 vertices. Independently, for the benchmark it checked all $6^8 = 1\,679\,616$ colour words of the consecutive edges of the transitive 9-vertex tournament against a chord-independent criterion implying the existence of a colour-avoiding 8-vertex path, with no failures.

Remark 4.6 (Novelty). The referee's literature search (September 2026) found no follow-up to [1] resolving Problem 1.1 for $q \geq 4$ and no prior occurrence of this example. This has not been checked by a human.

A Full proofs

Lemma 2.1. $f_{6,5}(9) = 8$.

Proof. Lower bound $f_{6,5}(9) \geq 8$. Let $v_1, \dots, v_9$ be the vertices of the transitive tournament in transitive order ($v_i \to v_j$ for $i < j$), let χ be any 6-edge-colouring, and put $a_i = \chi(v_i v_{i+1})$ for $1 \leq i \leq 8$. In a transitive tournament every directed path on a set of 8 vertices is the increasing order of that set, so the 8-vertex paths are exactly the nine paths obtained by deleting one vertex from $v_1 v_2 \cdots v_9$. Suppose, for a contradiction, that none of them is colour-avoiding, i.e. each uses all six colours.

Deleting v_9 gives the path $v_1 \cdots v_8$ with edge colours $a_1 \cdots a_7$; deleting v_1 gives $v_2 \cdots v_9$ with colours $a_2 \cdots a_8$. Both words contain all six colours, hence so does $a_1 \cdots a_8$. Call a position $i \in \{1, \dots, 8\}$ a *singleton position* if the colour a_i occurs exactly once in the word $a_1 \cdots a_8$. If r colours occur exactly once, the other $6 - r$ colours occur at least twice, so $8 \geq r + 2(6 - r) = 12 - r$ and $r \geq 4$: there are at least four singleton positions.

Position 1 is not singleton: otherwise the path $v_2 \cdots v_9$, whose colours are $a_2 \cdots a_8$, misses the colour a_1. Symmetrically position 8 is not singleton.

Two singleton positions i and $i + 1$ cannot both be singleton. Indeed, then $a_i \neq a_{i+1}$ (each occurs once) and $1 \leq i \leq 7$; delete v_{i+1}. The resulting 8-vertex path $v_1 \cdots v_i v_{i+2} \cdots v_9$ uses the edges of colours a_j, $j \notin \{i, i+1\}$, together with the single chord $v_i v_{i+2}$. The colours a_i and a_{i+1} occur nowhere else in $a_1 \cdots a_8$, and the one chord carries one colour, so at least one of a_i, a_{i+1} is missing from this path, a contradiction.

Hence at least four singleton positions lie in $\{2, \dots, 7\}$ and are pairwise non-consecutive. But a set of pairwise non-consecutive integers in an interval of six consecutive integers has at most three elements. This contradiction shows that every 6-edge-colouring of the transitive 9-vertex tournament has a colour-avoiding path on 8 vertices, i.e. $f_{6,5}(9) \geq 8$.

Upper bound $f_{6,5}(9) \leq 8$. Colour the eight consecutive edges $v_1 v_2, \dots, v_8 v_9$ with $1, 2, 3, 4, 5, 6, 1, 2$ respectively and the remaining edges arbitrarily. The only directed path on 9 vertices of a transitive tournament is $v_1 \cdots v_9$, and its edges use all six colours, so it is not colour-avoiding. Hence $p(\chi) \leq 8$ for this colouring and $f_{6,5}(9) \leq 8$. □

Lemma 3.1. *Every directed path of T on 8 or more vertices uses all six colours, and the path $\ell_1 \ell_2 \ell_3 x_0 x_1 x_2 r_1$ has edge colours $1, 3, 6, 2, 5, 6$. Hence $p(T) = 7$.*

Proof. Structure of long paths. Every edge between different blocks points from L to C, from C to R, or from L to R. Hence a directed path visits vertices of L first, then vertices of C, then vertices of R (some of these groups possibly empty). Within L and within R the order is the transitive one; within C, a path through all three vertices is a cyclic rotation x_i, x_{i+1}, x_{i+2} for some $i \in \mathbb{Z}/3\mathbb{Z}$, and a path through exactly two vertices x_j, x_{j+1} traverses the edge $x_j \to x_{j+1}$ (the vertex x_{j+2} being omitted). A directed path on 8 vertices therefore omits exactly one of the nine vertices, and is determined by the omitted vertex together with, if all of C is present, the rotation i. (There are thus $3 + 6 \cdot 3 = 21$ such paths; this count is the referee's, Appendix B, and is not needed below.)

The colours of D. Suppose an 8-vertex path contains ℓ_3, all of C in the order x_i, x_{i+1}, x_{i+2}, and r_1. Its four edges $\ell_3 x_i$, $x_i x_{i+1}$, $x_{i+1} x_{i+2}$, $x_{i+2} r_1$ have colours $d_{i-1}, d_i, d_{i+1}, d_{i+2}$, which include all three residues, i.e. all of $D = \{2, 5, 6\}$. If instead the omitted vertex is $x_j \in C$, the path contains

$\ell_3, x_{j+1}, x_{j+2}, r_1$ consecutively; writing $i = j + 1$, the edges $\ell_3 x_i$, $x_i x_{i+1}$, $x_{i+1} r_1$ have colours d_{i-1}, d_i, d_{i+1}, again all of D.

The colours $1, 3, 4$, *and* D *in the remaining cases.* We go through the nine possible omitted vertices.

- x_j omitted ($j \in \mathbb{Z}/3\mathbb{Z}$): the path is $\ell_1 \ell_2 \ell_3 x_{j+1} x_{j+2} r_1 r_2 r_3$. The L-part contributes $\chi(\ell_1 \ell_2) = 1$, $\chi(\ell_2 \ell_3) = 3$; the R-part contributes $\chi(r_1 r_2) = 4$, $\chi(r_2 r_3) = 1$; D is covered by the previous paragraph.
- ℓ_1 omitted: $\ell_2 \ell_3 x_i x_{i+1} x_{i+2} r_1 r_2 r_3$. Colour 3 from $\ell_2 \ell_3$; 4 and 1 from $r_1 r_2$, $r_2 r_3$; D from the previous paragraph.
- ℓ_2 omitted: $\ell_1 \ell_3 x_i x_{i+1} x_{i+2} r_1 r_2 r_3$. The chord $\ell_1 \ell_3$ has colour 3; R gives $4, 1$; D as before.
- ℓ_3 omitted: $\ell_1 \ell_2 x_i x_{i+1} x_{i+2} r_1 r_2 r_3$. Colour 1 from $\ell_1 \ell_2$, colour 3 from the new boundary edge $\ell_2 x_i$, colours $4, 1$ from R. The edges $x_i x_{i+1}$, $x_{i+1} x_{i+2}$, $x_{i+2} r_1$ have colours d_i, d_{i+1}, d_{i+2}, all of D.
- r_1 omitted: $\ell_1 \ell_2 \ell_3 x_i x_{i+1} x_{i+2} r_2 r_3$. L gives $1, 3$; the new boundary edge $x_{i+2} r_2$ has colour 4; $r_2 r_3$ has colour 1. The edges $\ell_3 x_i$, $x_i x_{i+1}$, $x_{i+1} x_{i+2}$ have colours d_{i-1}, d_i, d_{i+1}, all of D.
- r_2 omitted: $\ell_1 \ell_2 \ell_3 x_i x_{i+1} x_{i+2} r_1 r_3$. L gives $1, 3$; the chord $r_1 r_3$ has colour 4; D from the previous paragraph.
- r_3 omitted: $\ell_1 \ell_2 \ell_3 x_i x_{i+1} x_{i+2} r_1 r_2$. L gives $1, 3$; $r_1 r_2$ has colour 4; D from the previous paragraph.

In every case the path uses $\{1, 3, 4\} \cup D = \{1, 2, 3, 4, 5, 6\}$. A directed path on 9 vertices contains a directed path on 8 vertices (its first eight vertices), whose edge colours form a subset of its own; so it too uses all six colours. Hence no directed path on 8 or more vertices is colour-avoiding, and $p(T) \le 7$.

The witness. The path $\ell_1 \to \ell_2 \to \ell_3 \to x_0 \to x_1 \to x_2 \to r_1$ is a directed path on 7 vertices with edge colours

$$\chi(\ell_1 \ell_2) = 1, \qquad \chi(\ell_2 \ell_3) = 3, \qquad \chi(\ell_3 x_0) = d_2 = 6,$$
$$\chi(x_0 x_1) = d_0 = 2, \qquad \chi(x_1 x_2) = d_1 = 5, \qquad \chi(x_2 r_1) = d_2 = 6,$$

which avoid colour 4. Hence $p(T) \ge 7$, and $p(T) = 7$. □

B Report of the LLM referee (verbatim)

> The following is the unedited report of the adversarial referee agent (`claude-fable-5`, 2026-09-02/03) on the *original* writeup. Its step labels (S0, L1–L7, C1–C8) refer to that writeup, not to this note. The scripts it mentions are in `verification/scripts/2512.10438__00/`. Treat it as machine output, not as an authority.

Target: `attacks/2512.10438__00/output.md`, claiming an affirmative resolution of Problem 5.1 of Fox–Sudakov–Wigderson, "Color-avoiding directed paths in tournaments" (arXiv:2512.10438v2), via an explicit 6-edge-colored non-transitive tournament on 9 vertices with longest color-avoiding path on 7 vertices, against a transitive benchmark f_{6,5}(9)=8.

B.1 Interpretation

I fetched arXiv:2512.10438v2 (PDF, 22 pp.) and compared verbatim.

- **Problem 5.1 (paper, p. 21, quoted from the PDF):** "For some integers q ≥ 4 and N, does there exist a non-transitive q-edge-colored N-vertex tournament whose longest color-avoiding path has strictly fewer than f_{q,q−1}(N) vertices?" This is exactly the statement the writeup addresses. The catalog page's transcription also matches.
- **Length convention (paper, p. 1):** "Throughout this paper, by the *length* of a path, we mean the number of vertices it comprises." The writeup counts vertices. Match.

- **Definition of f (paper, p. 2):** $g_{q,r}(N)$ is "the minimum, over all q-edge-colored N-vertex tournaments, of the number of vertices in the longest directed path whose edges receive at most r colors"; $f_{q,r}(N)$ is defined "identically ... except restricting our attention only to q-edge-colorings of the N-vertex transitive tournament." The writeup's $f_{q,q-1}(N) = \min_\chi p(\chi)$ over transitive colorings, with p = max vertices of a path using at most q−1 colors, is exactly this. Match.
- **Spirit of the problem.** The problem explicitly asks for "*some* integers $q \geq 4$ and N", so a single finite instance resolves it as posed. This is not a strawman reading: the surrounding text (p. 7 and p. 21) says the authors "do not expect $f_{q,q-1}(N) = g_{q,q-1}(N)$" and "do not expect [the extremal construction] to be precisely transitive" — an affirmative answer is exactly the direction they anticipated, and the q=3 precedent they cite (Hamaker–Stein, giving $G_{3,2}(7) \geq 19 > 17 = F_{3,2}(7)$) is itself a finite instance. A reasonable author of the paper would consider the literal Problem 5.1 resolved by this example, while noting that the deeper asymptotic question (behavior of $g_{q,q-1}$ vs $f_{q,q-1}$ as $N \to \infty$, e.g. for the exponent) remains open. The writeup claims only the literal problem.
- The example is non-transitive (contains the 3-cycle $x0 \to x1 \to x2 \to x0$) but not strongly connected; Problem 5.1 imposes no connectivity condition, and the writeup flags this itself.

Interpretation: fair and intended. `interpretation_ok: true`.

B.2 Step-by-step findings

step	label	note
S0. Definitions of $p(\chi)$ and $f_{q,q-1}(N)$	VALID	Matches the paper's definitions verbatim (vertices convention; min over transitive colorings).
L1. Under "no color-avoiding 8-vertex path", the words $a_1 \ldots a_7$ and $a_2 \ldots a_8$ each contain all 6 colors	VALID	Deleting v_9 (resp. v_1) gives an 8-vertex path using exactly edges $a_1 \ldots a_7$ (resp. $a_2 \ldots a_8$) and no chord; by assumption it uses all 6 colors. In a transitive tournament all 8-vertex paths are exactly the 9 vertex-deleted increasing subsequences.
L2. At least 4 singleton positions in $a_1 \ldots a_8$	VALID	All 6 colors occur (L1); if r colors occur once, $8 \geq r + 2(6-r) = 12 - r$, so $r \geq 4$. Re-derived; correct.
L3. Positions 1 and 8 are not singleton	VALID	If a_1 singleton, the path $v_2 \ldots v_9$ uses only $a_2 \ldots a_8$, missing color a_1 — contradiction; symmetric for a_8.

step	label	note
L4. No two consecutive singleton positions	VALID	If a_i, a_{i+1} singleton (necessarily distinct colors), deleting v_{i+1} removes both and adds one chord $v_i v_{i+2}$; a single chord color restores at most one of the two, so the 8-vertex path misses a color — contradiction. Chord exists since $2 \le i+1 \le 8$.
L5. Contradiction: 4 pairwise non-consecutive positions among {2,…,7}	VALID	Max independent set in a 6-term interval under non-consecutivity is 3 (e.g. {2,4,6}). Hence $f_{6,5}(9) \ge 8$.
L6. Upper bound $f_{6,5}(9) \le 8$ via word 1,2,3,4,5,6,1,2	VALID	The unique 9-vertex path of a transitive tournament is $v_1 \dots v_9$; its edges use all 6 colors, so no 9-vertex color-avoiding path exists for this coloring, regardless of the "arbitrary" chord colors.
L7. Conclusion $f_{6,5}(9) = 8$	VALID	L1–L6; independently machine-verified exhaustively (see below).
C1. The construction is a well-defined non-transitive tournament using all 6 colors	VALID	Machine-checked: 36 edges, one per pair; contains a directed triangle; colors {1,…,6} all occur.
C2. Every path meeting several blocks traverses L ⟶ C ⟶ R; 8-vertex paths are vertex-deleted orders, unique up to the C-rotation	VALID	All inter-block edges point L⟶C, C⟶R, L⟶R; within-block orders forced (L, R transitive; C-part is a rotation $x_i x_{i+1} x_{i+2}$ or a single edge $x_{j+1} x_{j+2}$). Exactly 21 eight-vertex paths exist (3 omitting a C-vertex, 6×3 omitting an L/R vertex); machine count agrees: 21.
C3. With all of C present and boundaries $\ell 3$, r1, the four colors d_{i-1}, d_i, d_{i+1}, d_{i+2} cover D = {2,5,6}	VALID	Four consecutive indices mod 3 cover all residues; $\ell 3 x_i = d_{i-1}$, $x_i x_{i+1} = d_i$, $x_{i+1} x_{i+2} = d_{i+1}$, $x_{i+2} r_1 = d_{i+2}$. Re-derived.
C4. With one C-vertex omitted, $\ell 3 x_i$, $x_i x_{i+1}$, $x_{i+1} r_1$ give d_{i-1}, d_i, d_{i+1} = D	VALID	Re-derived (here i = j+1 for omitted x_j).

step	label	note
C5. Case analysis for colors {1,3,4} over the 9 possible omitted vertices (incl. modified boundaries ℓ2x_i color 3 and x_i r_2 color 4, chords ℓ1ℓ3 color 3, r1r3 color 4)	VALID	Re-derived each of the 7 bullets by hand and confirmed by exhaustive machine check: all 21 eight-vertex paths use all six colors.
C6. Any longer path contains a contiguous 8-vertex subpath, hence uses all 6 colors; so p(T) ≤ 7	VALID	The first 8 vertices of a 9-vertex path form a directed path; a superset of an all-6-colors edge set uses all 6 colors.
C7. Witness path ℓ1ℓ2ℓ3x0x1x2r1 has colors 1,3,6,2,5,6, avoiding color 4; p(T) = 7	VALID	Machine-checked: colors exactly [1,3,6,2,5,6]; exhaustive DFS over all directed paths gives longest color-avoiding path = 7 vertices.
C8. Conclusion: p(T) = 7 < 8 = f_{6,5}(9), answering Problem 5.1 affirmatively with (q,N) = (6,9)	VALID	Follows from L7 and C7; matches Problem 5.1 verbatim.

No step is a GAP or ERROR.

B.3 Reference check

The writeup is self-contained (no external theorems are invoked in the proof). I verified every factual claim about the source paper against the fetched PDF of arXiv:2512.10438v2:

- **Problem 5.1** exists on p. 21 and reads exactly as used (quoted above under Interpretation). Confirmed.
- **Definitions of f_{q,r}, g_{q,r} and the length-in-vertices convention** (pp. 1–2). Confirmed; the writeup's formalization is equivalent.
- **Proposition 1.5** (p. 6): "Let q ≥ r be integers with r > 2q/3. Then F_{q,r}(n) = G_{q,r}(n) for all n." This concerns the vector setting ((q−1)-comparable sets vs (q−1)-increasing sequences) and, per the paper's remark after Problem 5.1, rules out vector-based constructions for q ≥ 4 — consistent with the catalog context. It does **not** constrain direct tournament constructions like the writeup's; the paper's one-way implication

  (3) (p. 5) goes from vector constructions to tournament upper bounds only, so there is no conflict between Proposition 1.5 and the claimed example.

- **q = 3 case** (p. 21): affirmative via Hamaker–Stein (G_{3,2}(7) ≥ 19 vs F_{3,2}(7) = 17). Confirmed as background; not load-bearing.
- **Novelty check:** web searches (September 2026) found no follow-up paper or announcement resolving Problem 5.1 for q ≥ 4, and no prior occurrence of this example. Not already known as far as I can determine.

`references_ok: true`.

B.4 Computational check

Scripts (reproducible) in `verification/scripts/2512.10438__00/`:

1. `check_construction.py` — encodes the 9-vertex tournament and coloring exactly as in the writeup's table (remaining edges color 1). Results:
   - 36 edges, exactly one direction per pair (valid tournament);
   - contains a directed triangle (non-transitive); all 6 colors used;
   - exhaustive DFS over **all** directed paths: longest color-avoiding path has **7** vertices (witness ℓ1ℓ2ℓ3x0x1x2r1, colors {1,2,3,5,6}, avoiding 4);
   - the writeup's witness path has edge colors exactly [1,3,6,2,5,6];
   - there are exactly **21** eight-vertex directed paths and **every one uses all 6 colors**. So p(T) = 7. PASSED.

2. `check_benchmark.py` — verifies f_{6,5}(9) = 8 for **all** colorings, not a sample. Key reduction (checked in the script's comments): in the transitive tournament on 9 vertices, every path on ≥ 8 vertices uses only consecutive edges a_1,...,a_8 and at most one 2-chord v_{k−1}v_{k+1}. For each of the 6^8 = 1,679,616 consecutive-edge words the script checks the chord-independent sufficient condition: (A) a_2...a_8 misses a color, or

   (B) a_1...a_7 misses a color, or (C) for some k ∈ {2,...,8} the word minus positions k−1, k misses ≥ 2 colors (so the single chord cannot restore both). **All 1,679,616 words satisfy A, B or C** (0 failures), proving that every 6-coloring of the transitive 9-vertex tournament — whatever the chord colors — has a color-avoiding 8-vertex path: f_{6,5}(9) ≥ 8. The word 1,2,3,4,5,6,1,2 puts all 6 colors on the unique 9-vertex path, giving f_{6,5}(9) ≤ 8. Hence **f_{6,5}(9) = 8**. PASSED.

Together these machine-verify the entire theorem: p(T) = 7 < 8 = f_{6,5}(9). `computation_run: true`.

B.5 Caveats

- **Not strongly connected.** The example's strong components are ordered ℓ1, ℓ2, ℓ3, {x0,x1,x2}, r1, r2, r3. Problem 5.1 requires only non-transitivity, which holds (one 3-cycle). The writeup discloses this.
- **Minimal non-transitivity.** The tournament differs from transitive by a single reversed edge's worth of structure (one 3-cycle); this is consistent with the paper's Theorem 1.2, which forces extremal examples to be o(1)-close to transitive, so no red flag arises from that direction.
- **Literal vs asymptotic.** The problem as posed asks for a single (q,N); it is resolved. The natural follow-up the authors would still care about — whether g_{q,q−1}(N) is asymptotically smaller than f_{q,q−1}(N) for q ≥ 4 (e.g. by a power of N, as for q = 3) — is *not* addressed by a one-point gap of 1 at (6,9). The writeup's verdict line claims only Problem 5.1, which is accurate.
- **Convention dependence.** All quantities are counted in vertices, matching the paper's stated convention; no off-by-one issue. Colorings need not be surjective in the paper's definition, but the construction uses all 6 colors, so nothing hinges on this.
- **Unspecified edges.** The writeup colors "remaining" edges with color 1 and claims they are never needed; the exhaustive path check confirms the conclusion holds with this completion (and the ≥8-vertex-path analysis is completion-independent, since such paths never use those edges except the L⟶R chords, which never lie on a path of ≥ 8 vertices here — confirmed by the enumeration of all 21 such paths).
- Degenerate cases (k = 0/1 colors, tiny N) do not arise: the claim is a single explicit instance plus an exact benchmark value, both fully checked.

B.6 Referee summary

I set out assuming the writeup was wrong and failed to break it. The interpretation is exactly Problem 5.1 of arXiv:2512.10438v2, quoted verbatim from the paper, in the paper's own vertex-count convention, and the authors' surrounding remarks show an affirmative finite example is precisely what was asked for. Every step of the transitive benchmark proof (f_{6,5}(9) = 8) is valid on re-derivation, and I additionally machine-verified it exhaustively over all 6^8 consecutive-edge words via a chord-independent criterion that covers every coloring. The explicit 9-vertex, 6-colored, non-transitive tournament was independently re-encoded and brute-forced: it is a valid non-transitive tournament, all 21 of its 8-vertex directed paths use all six colors, and its longest color-avoiding path has exactly 7 vertices, avoiding color 4. Hence p(T) = 7 < 8 = f_{6,5}(9), and Problem 5.1 is answered affirmatively with (q,N) = (6,9). No prior or follow-up resolution was found in the literature. Verdict: CONFIRMED, high confidence.

References

[1] J. Fox, B. Sudakov and Y. Wigderson, Color-avoiding directed paths in tournaments, arXiv:2512.10438 (2025; v2 January 2026).